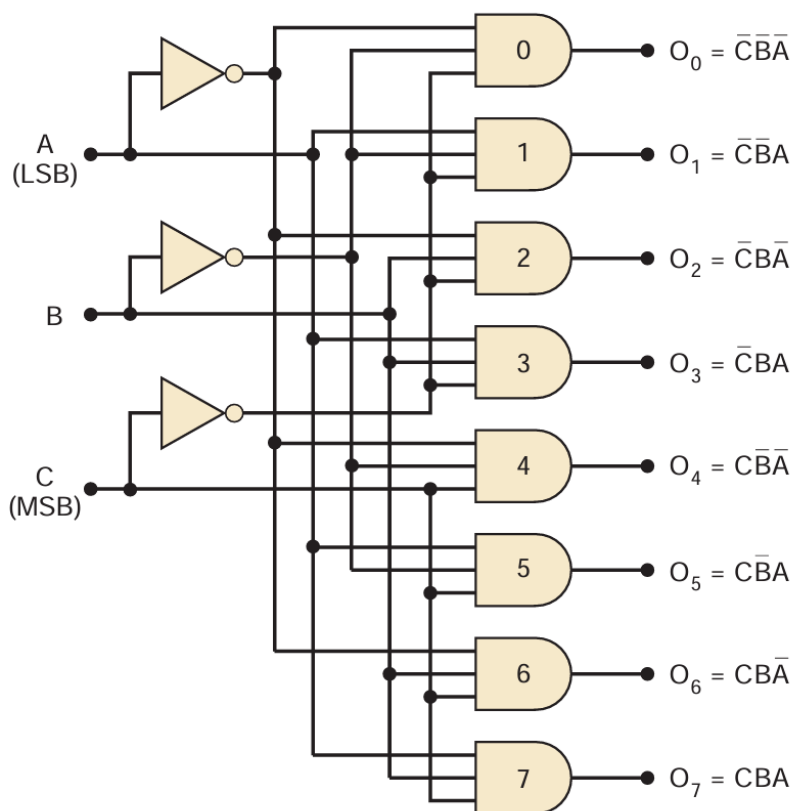


Design: 3-to-8-line Decoder



C	B	A	O_7	O_6	O_5	O_4	O_3	O_2	O_1	O_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

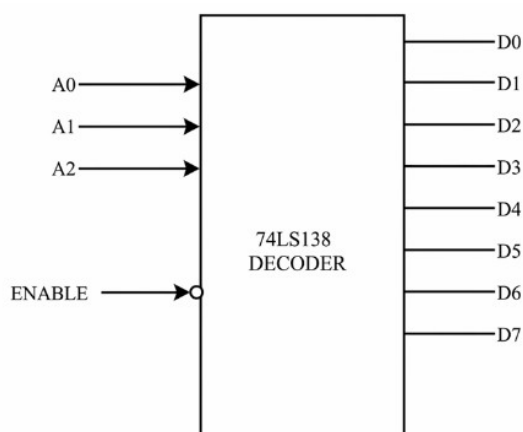


Fig - 3X8 Decoder

```

module decoder_3to8 (
input wire [2:0] sel,
output wire [7:0] y);

wire A,B,C;

                                sel
assign A = sel[0];              000 000
assign B = sel[1];              001 000
assign C = sel[2];              010 000
                                011 000
assign y[0] = ((~C)&(~B)&A);     100 000
assign y[1] = ((~C)&(~B)&~A);    101 000
assign y[2] = ((~C)&(B)&A);      110 010
assign y[3] = ((~C)&(B)&~A);     111 100
assign y[4] = ((C)&(~B)&A);
assign y[5] = ((C)&(~B)&~A);
assign y[6] = ((C)&(B)&A);
assign y[7] = ((C)&(B)&~A);

endmodule

```

```

`timescale 1ns/1ps

module tb_decoder_3to8;

    reg [2:0] sel;
    wire [7:0] y;
    integer i;

    decoder_3to8 uut (
        .sel(sel),
        .y(y)
    );

    initial begin
        $display("sel      y");
        $monitor("%b %b", sel, y);
        for (i = 0; i < 8; i = i + 1) begin
            sel = i;
            #10;
        end

        $finish;
    end

endmodule

```